

## 01 Sets (Set Concepts)



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How many stones are in my hand?

Think alone

Pair talk

Share evidence

Formula / rule

Worked step

Your turn

Teacher note: ask for first ideas; do not give the answer yet.

## Today's mission



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**I can distinguish finite and infinite sets.**

**1**

stones, bottle caps, sand, and set loops.

**2**

explain one example with a clear reason.

**3**

count, compare, define, explain.

## Starter: think first



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Which collections  
can we finish  
counting?

compare stones,  
bottle caps, and  
sand.

say what makes a  
set finite.

Formula / rule

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Worked step

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Your turn

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## Can We Count Everything?



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**A finite set has a last member; an infinite set keeps going.**

**Formula / rule**

**Worked step**

**Your turn**

**1**

**Stones in a bowl can be counted and finished.**

**2**

**Counting numbers keep going without a last number.**

## Defining Our Sets



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Worked example:

Formula / rule

Worked step

Your turn

1

**Finite set:** bottle caps on one desk have an end.

2

**Infinite set:** counting numbers continue forever.



## Try it together



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**Answer with evidence:**

? Which collection has a count you can finish?

? Which example shows the idea of no last member?

? Explain using finite or infinite.

## Activity challenge



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Count ten bottle caps and draw the finite

Draw a second set pattern that continues

Write one sentence comparing the two sets.

Formula / rule

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Worked step

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Your turn

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## Exit ticket



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**Can a very large set still be finite?**

**EXIT TICKET**

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**Use one complete sentence.**

**Use one key word: finite or infinite.**

**Hand in or read your answer aloud.**